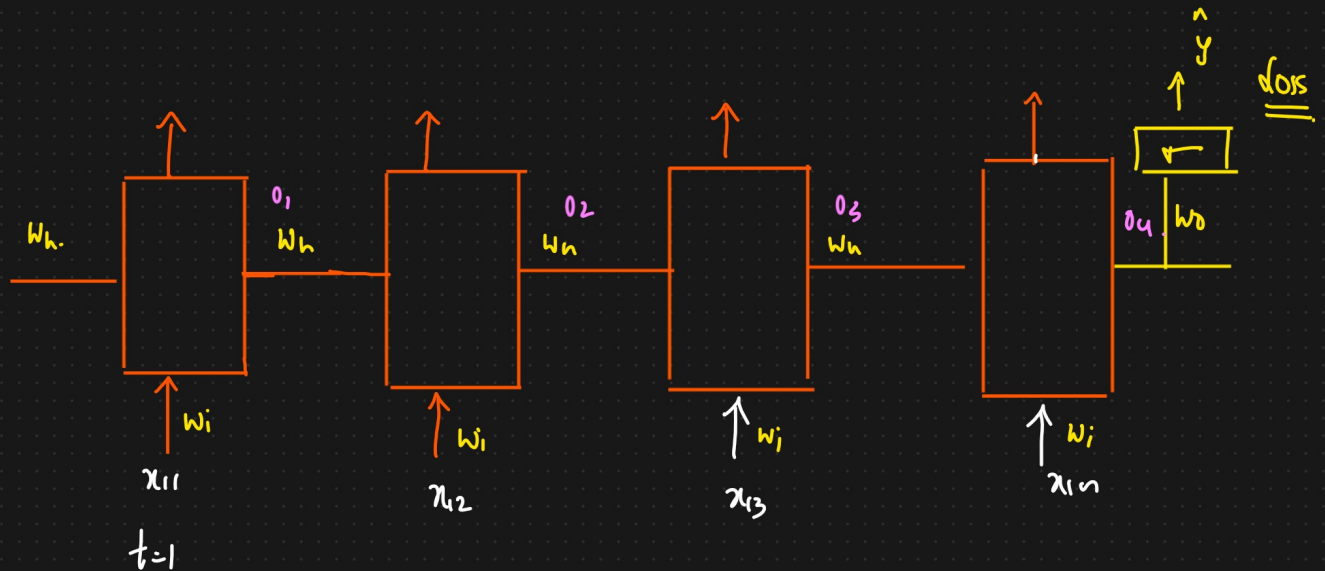


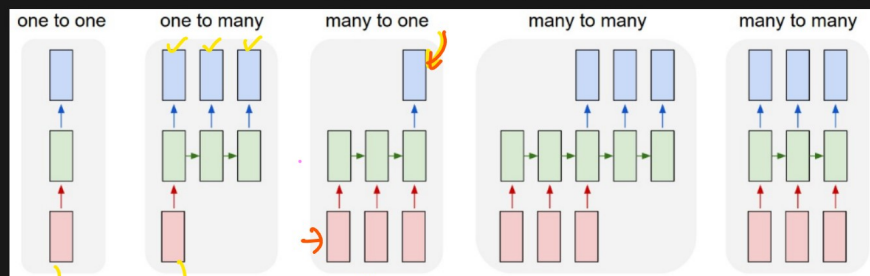
# Bidirectional RNN

- ① Simple RNN  $\rightarrow$  practical Implement  $\rightarrow$  Embedding layers
- ② LSTM, GRU Variants RNN  $\rightarrow$  Practical Implement
- ③ Bidirectional RNN / LSTM RNN



## Types of RNN

- ① One to Many RNN
- ② Many to One RNN
- ③ Many to Many RNN
- ④ One to One RNN



one i/p  
one o/p

one i/p  
o/p many

Image search  
I/p many  
o/p one

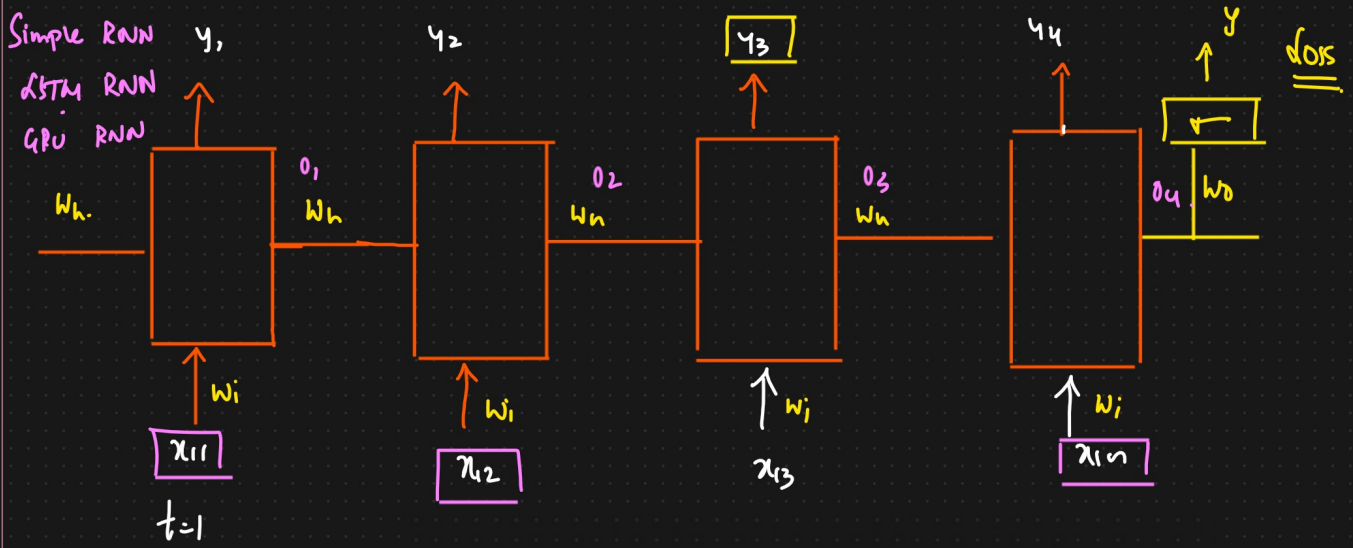
A cat eating  
food  
o eating

Language Translation  
many I/p  
many o/p

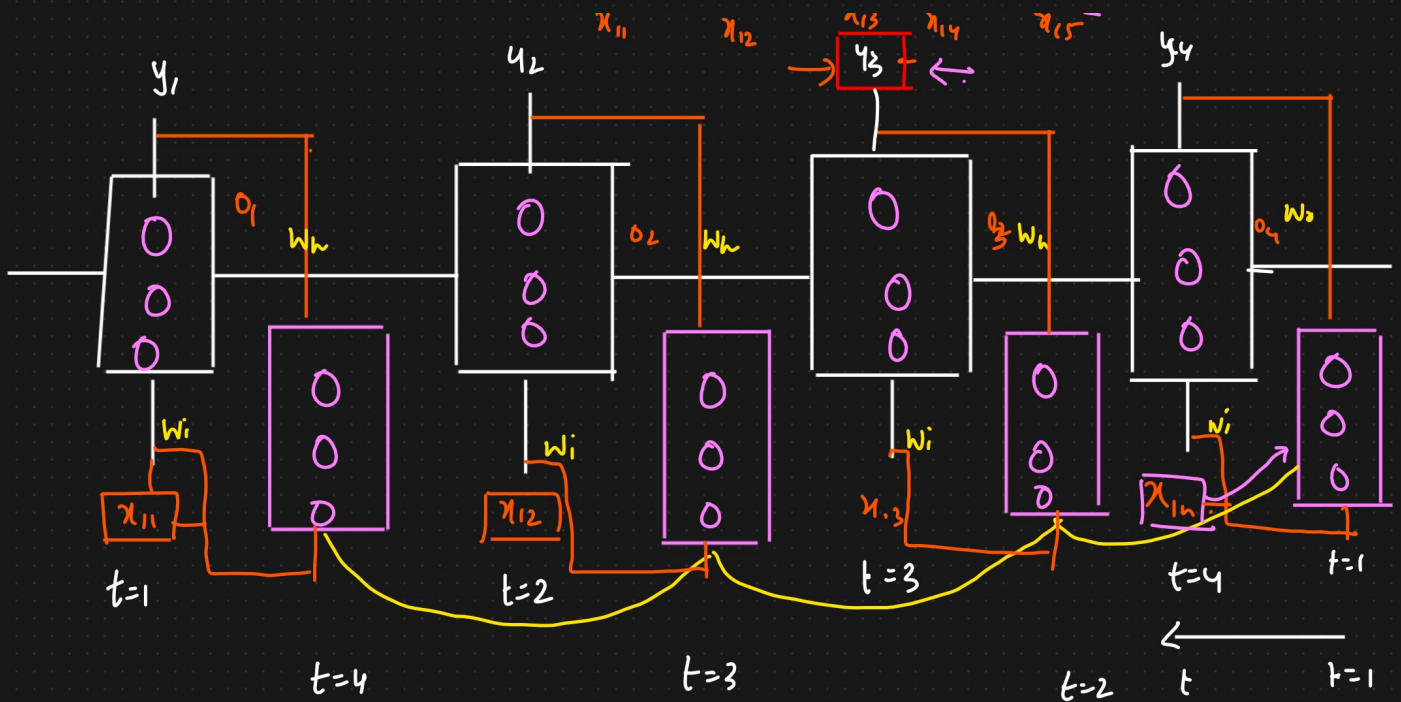
Eg:

Eg: Image  
Captioning

There is a dog and cat



## Bidirectional RNN



① Forward propagation  $\rightarrow$  Equation  $\leftarrow$